CMAI 480

Computer Vision

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Feature Extraction and Matching

Chapter 7 – Section 7.1.1

Reading

- Szeliski: 7.1.1

Feature Extraction



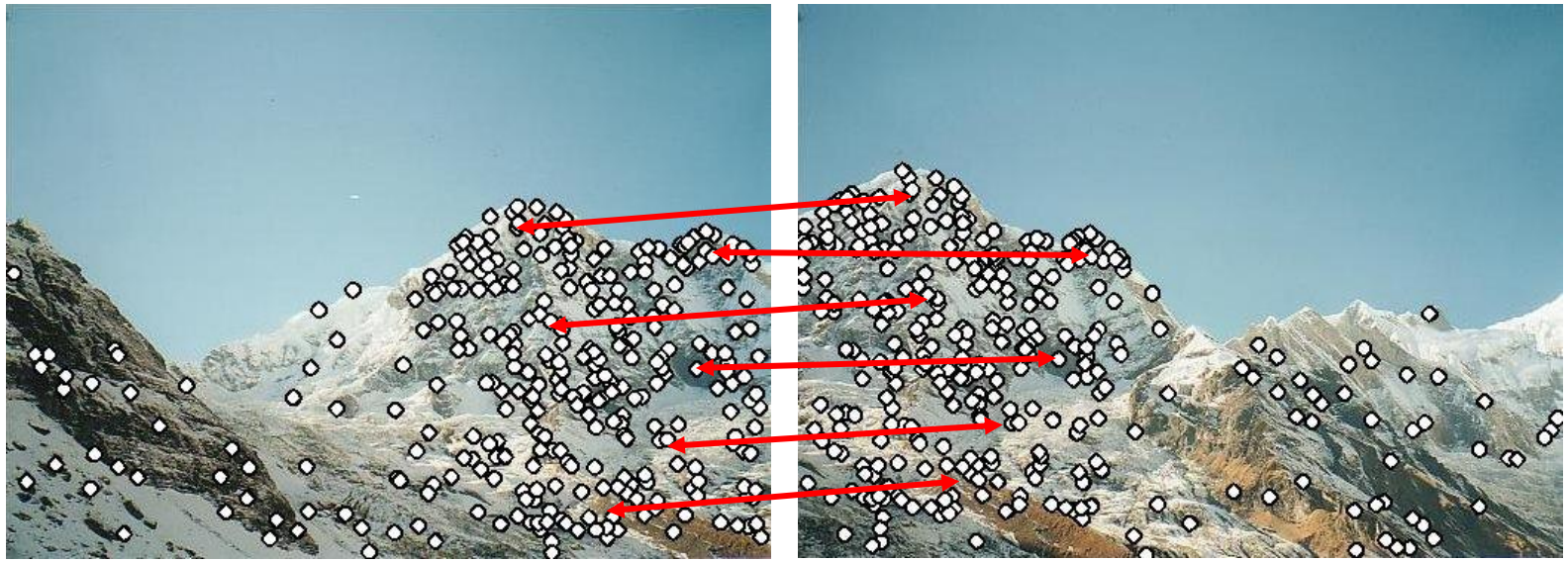
Why extract features?

- **Motivation**: panorama stitching (Next Chapter or Class Topic)
  - We have two images – how do we combine them?




Why extract features?

- Motivation: panorama stitching
  - We have two images – how do we combine them?



Step 1: extract features
Step 2: match features

Why Extract Features?

- Motivation: panorama stitching
  - We have two images – how do we combine them?



Overlapped region is a must

Step 1: extract features

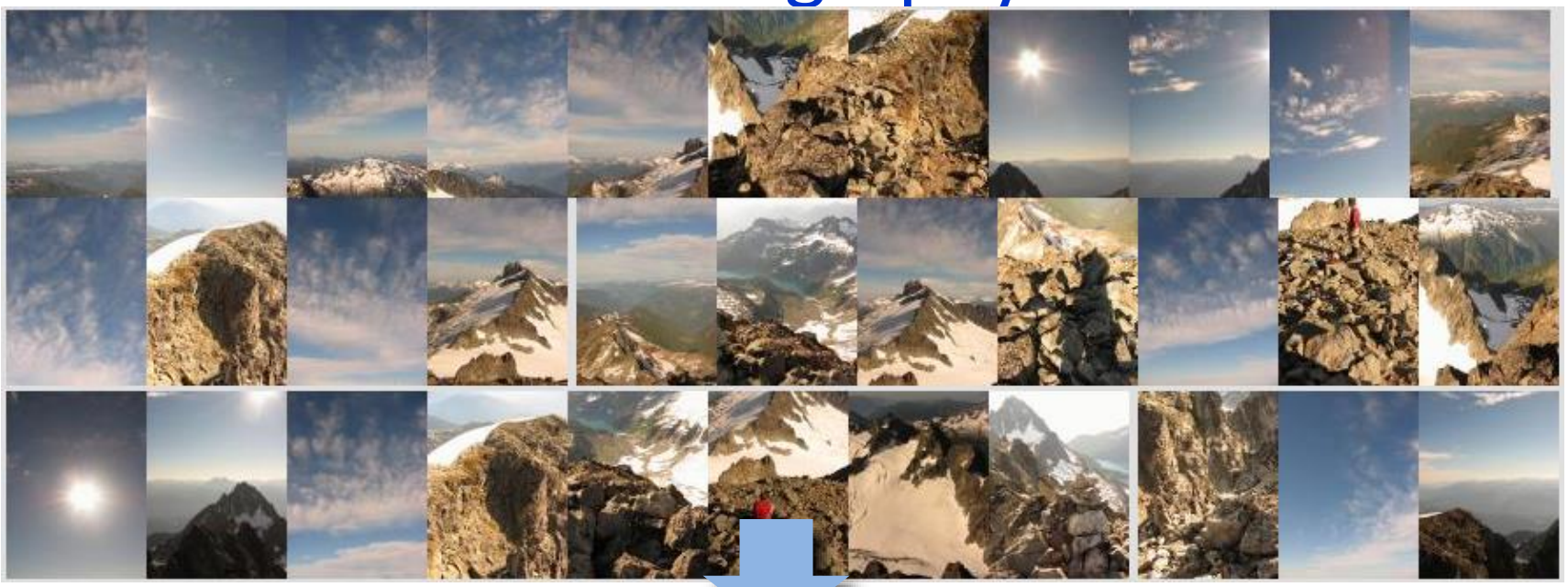
Step 2: match features

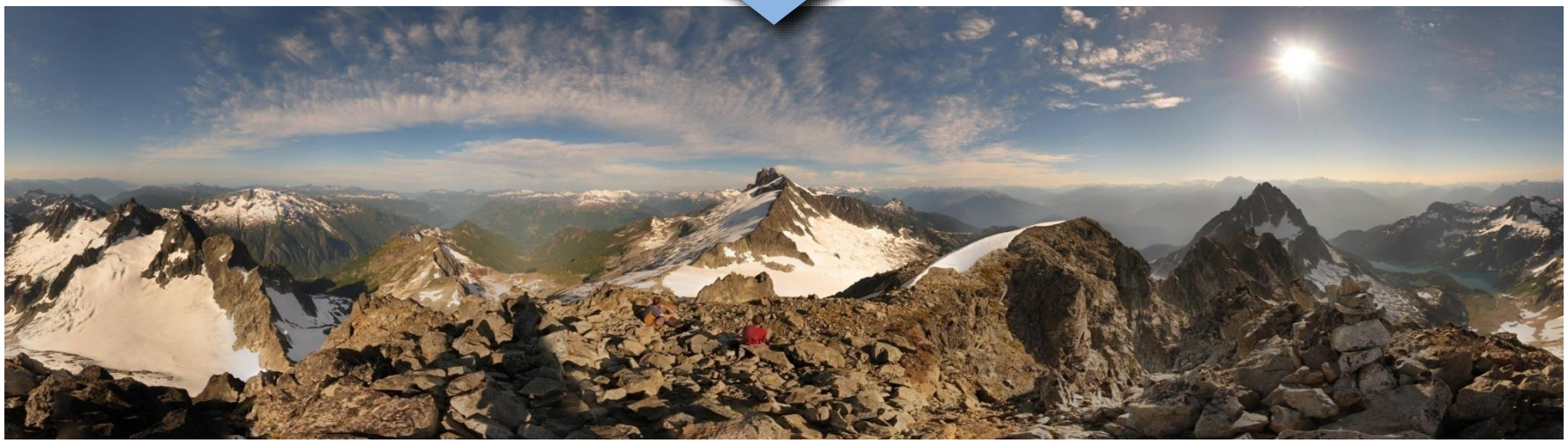
Step 3: Automatic Homography Estimation

Step 4: align images

Motivation: Automatic Panoramas

Autmoatic Homography Estimation

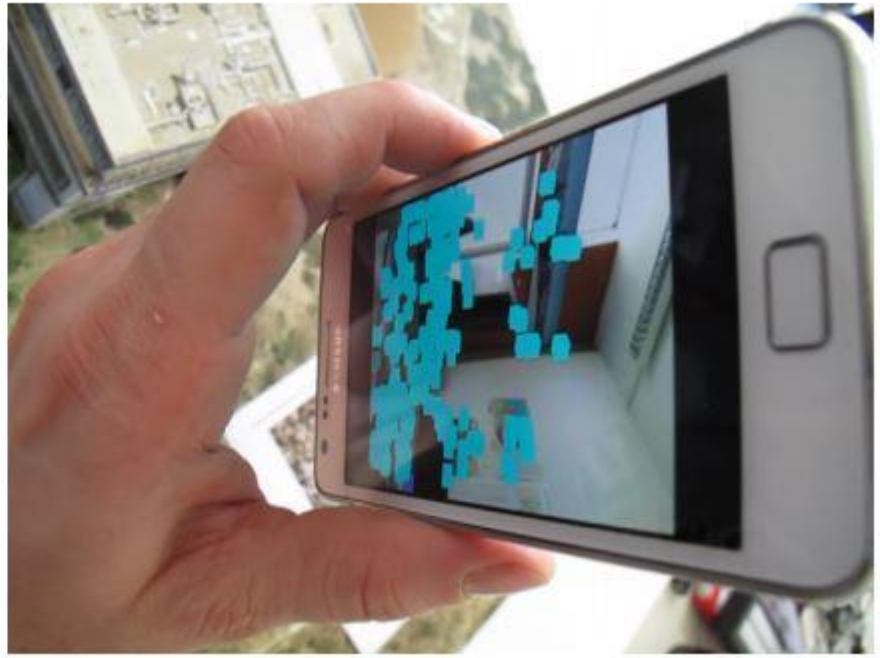




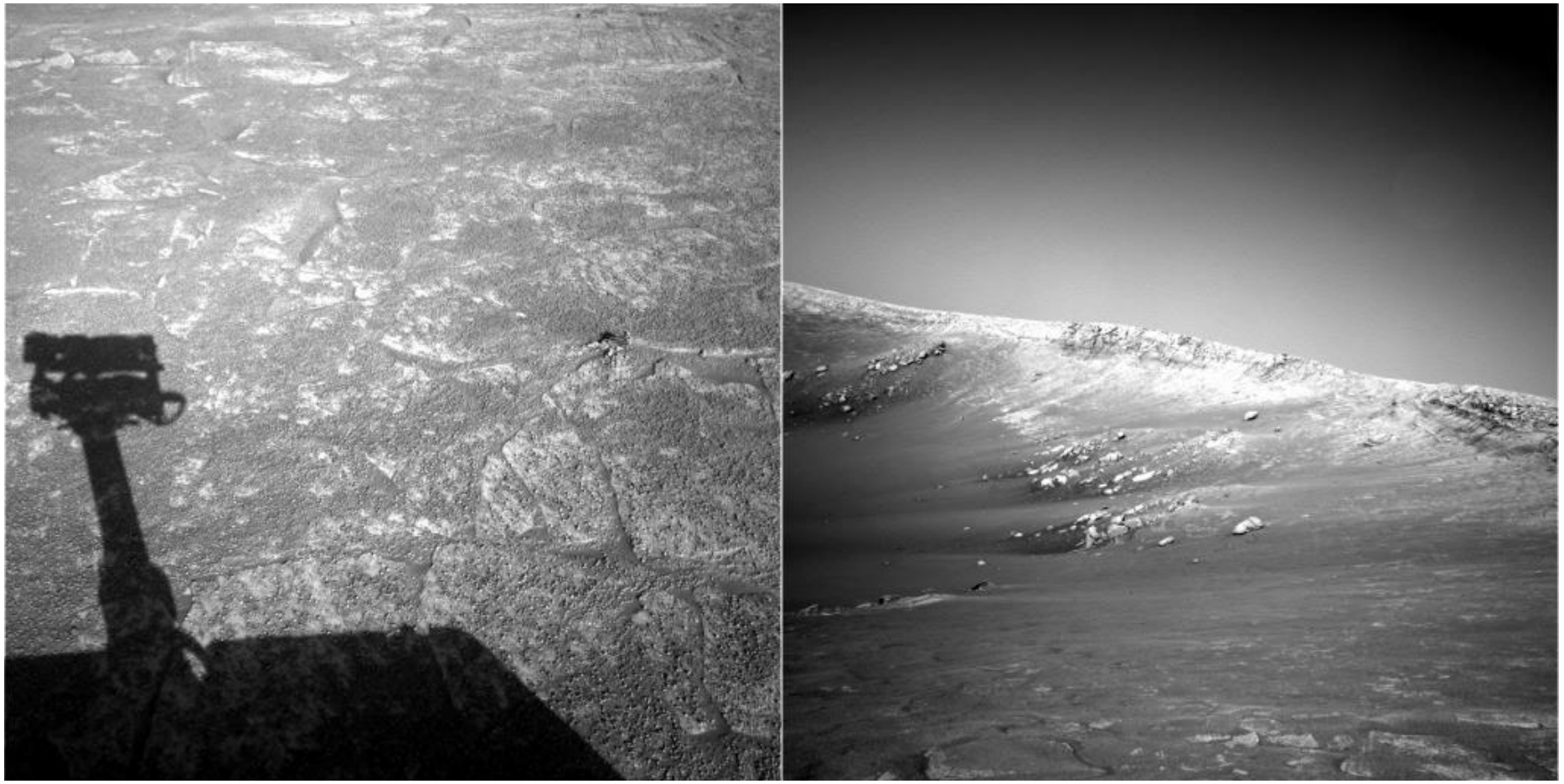
Application: Visual SLAM

- (aka Simultaneous Localization and Mapping)

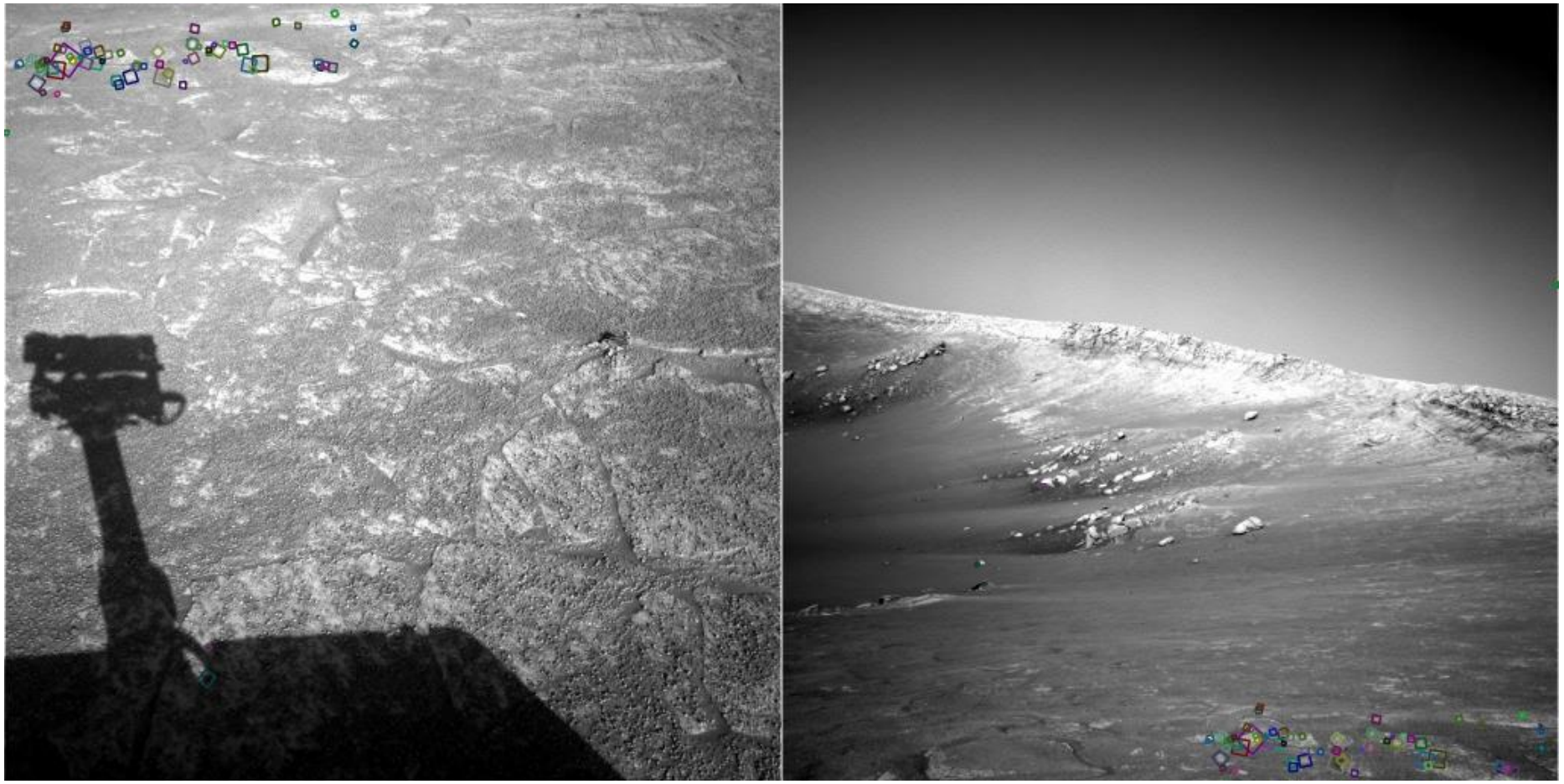




Do these images overlap?

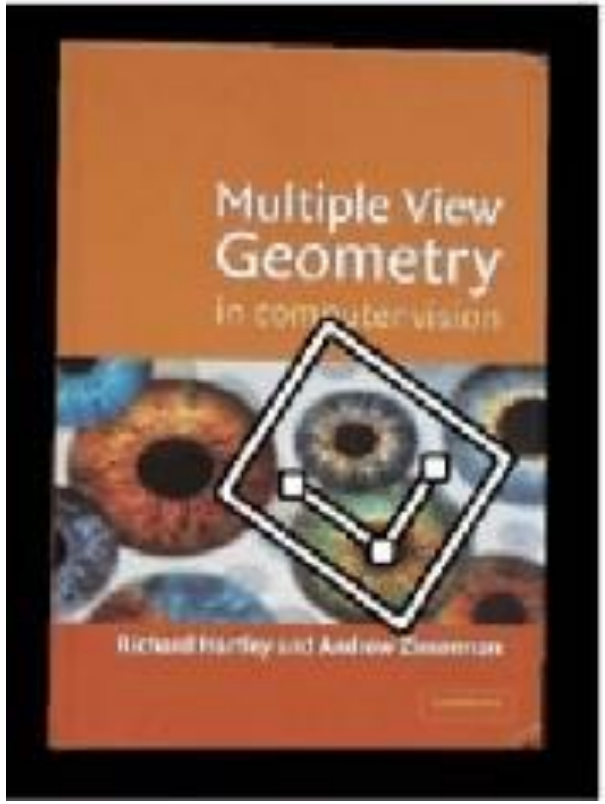


Answer below (look for tiny colored squares…)



NASA Mars Rover images
with **SIFT feature** matches

Another Application:
Feature matching for object search





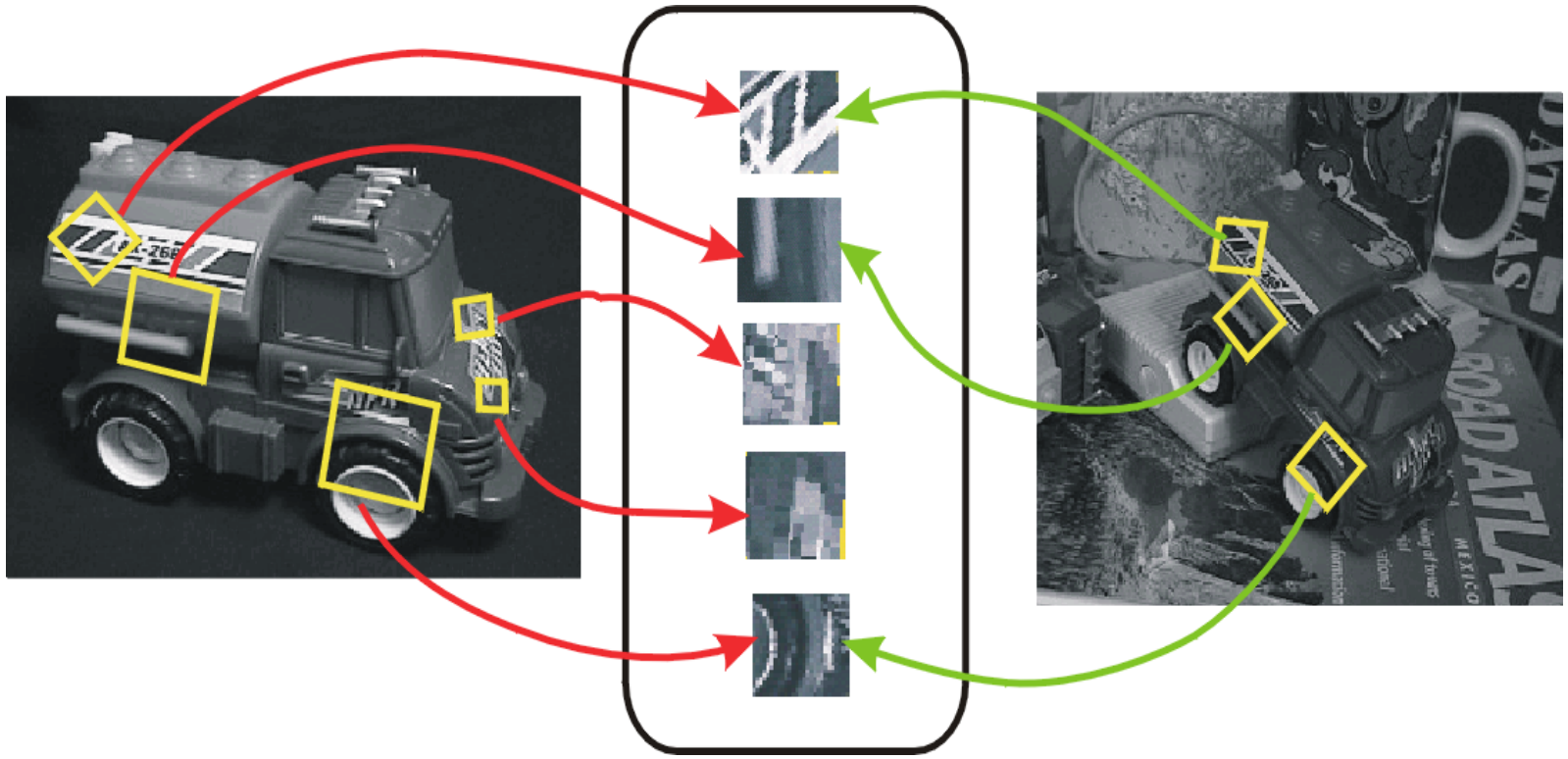
Feature matching



Invariant local features

Find features that are invariant to transformations

- geometric invariance: translation, rotation, scale
- photometric invariance: brightness, exposure, …

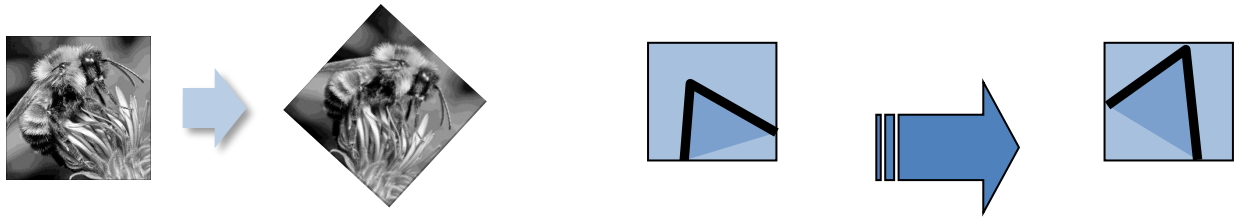


Feature Descriptors

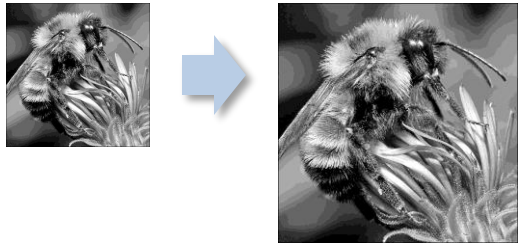
Image transformations

- Geometric

Rotation

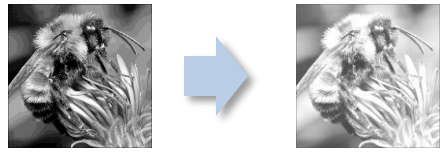


Scale



- Photometric

Intensity change



Advantages of local features

Locality

- features are local, so robust to occlusion and clutter

Quantity

- hundreds or thousands in a single image

Distinctiveness:

- can differentiate a large database of objects

Efficiency

- real-time performance achievable

More motivation…

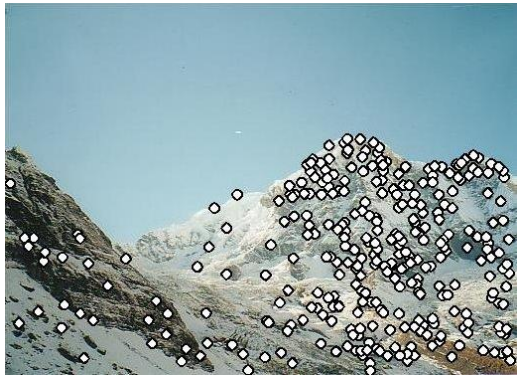
Feature points are used for:

- Image alignment (e.g., mosaics)
- 3D reconstruction
- Motion tracking (e.g. for AR)
- Object recognition
- Image retrieval
- Robot/car navigation
- … other

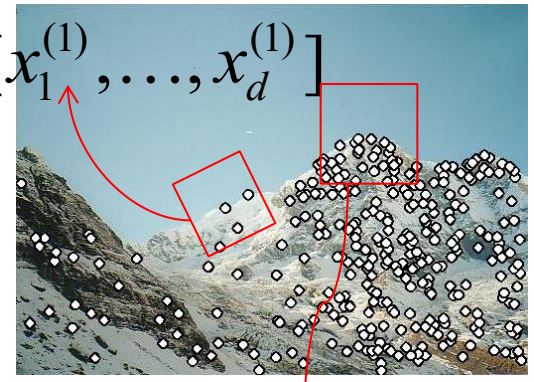


Local features: main components

1) **Detection**: Identify the interest points

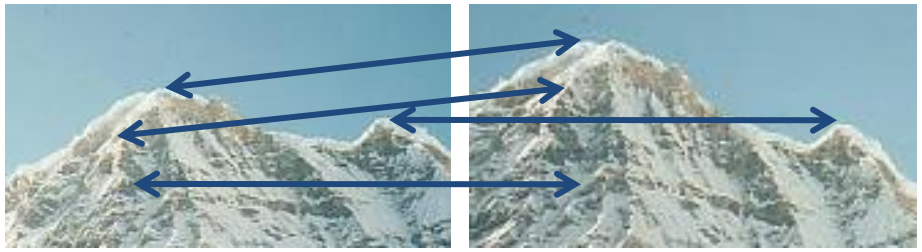


2) **Description**: Extract vector feature descriptor surrounding each interest point



$$\mathbf{x}_2 = [x_1^{(2)}, \ldots, x_d^{(2)}]$$

3) **Matching**: Determine correspondence between descriptors in two views



What makes a good feature?

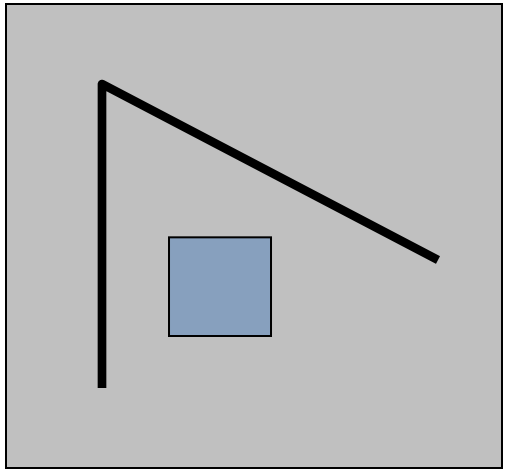
Uniqueness

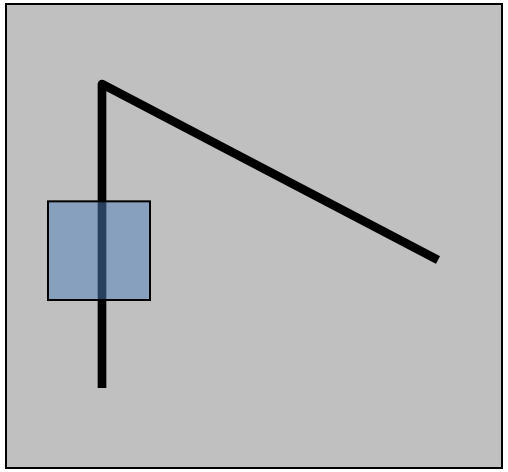


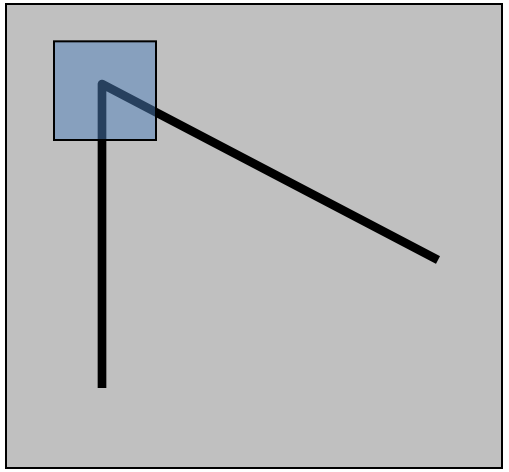
Local measures of uniqueness

Suppose we only consider a small window of pixels

- What defines whether a feature is a good or bad candidate?





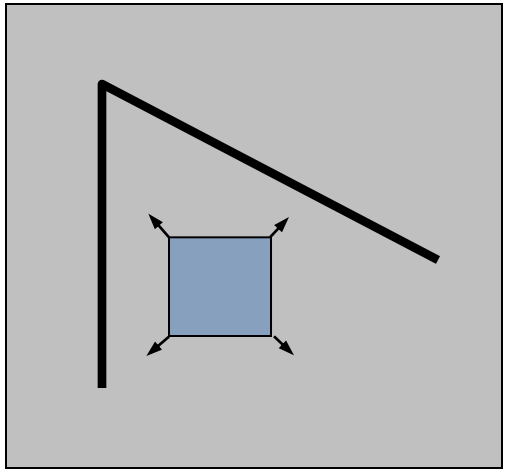


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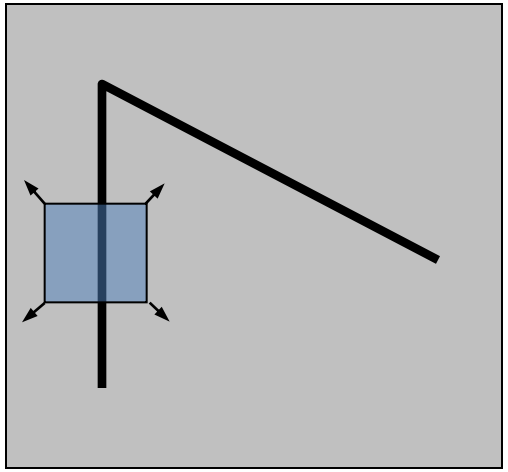
We want to find windows that contain strong gradients AND gradients oriented in more than one direction

Local measures of uniqueness

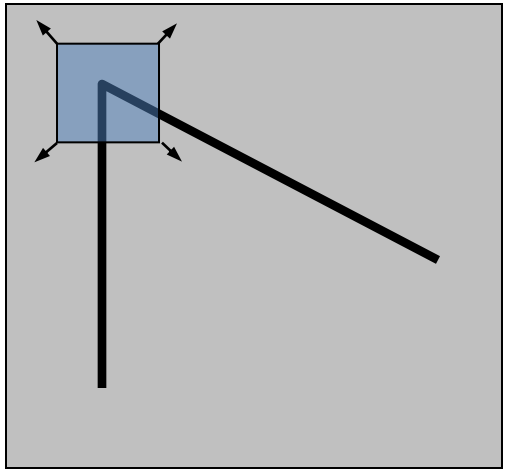
- How does the window change when you shift it?
- Shifting the window in any direction causes a big change



"flat" region:
no change in all directions



"edge":
no change along the edge direction



"corner":
significant change in all directions

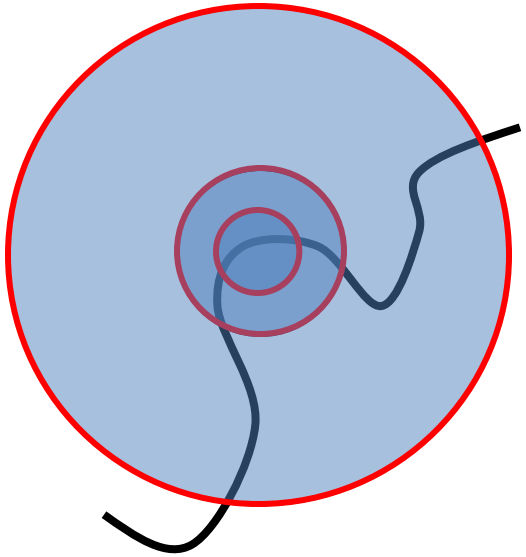
How to extract Robust Features/Corners with Invariant Properties? Gradients?

- SIFT Features
- Harris Corners



Fast Forward: Scale invariant detection

Suppose you're looking for corners



Key idea: find scale that gives local maximum of *f*

- in both position and scale
- One definition of *f*: the Harris operator

Upcoming Important Topics to cover

- Gradients
- The *Laplacian of Gaussian* (*LoG*)
- Automatic Selection of Scale
  - Gaussian pyramid
- SIFT Features

Questions